# VLANS IMPLEMANTATION
# scenario-[3]

**Designed, implemented,tested & documented**
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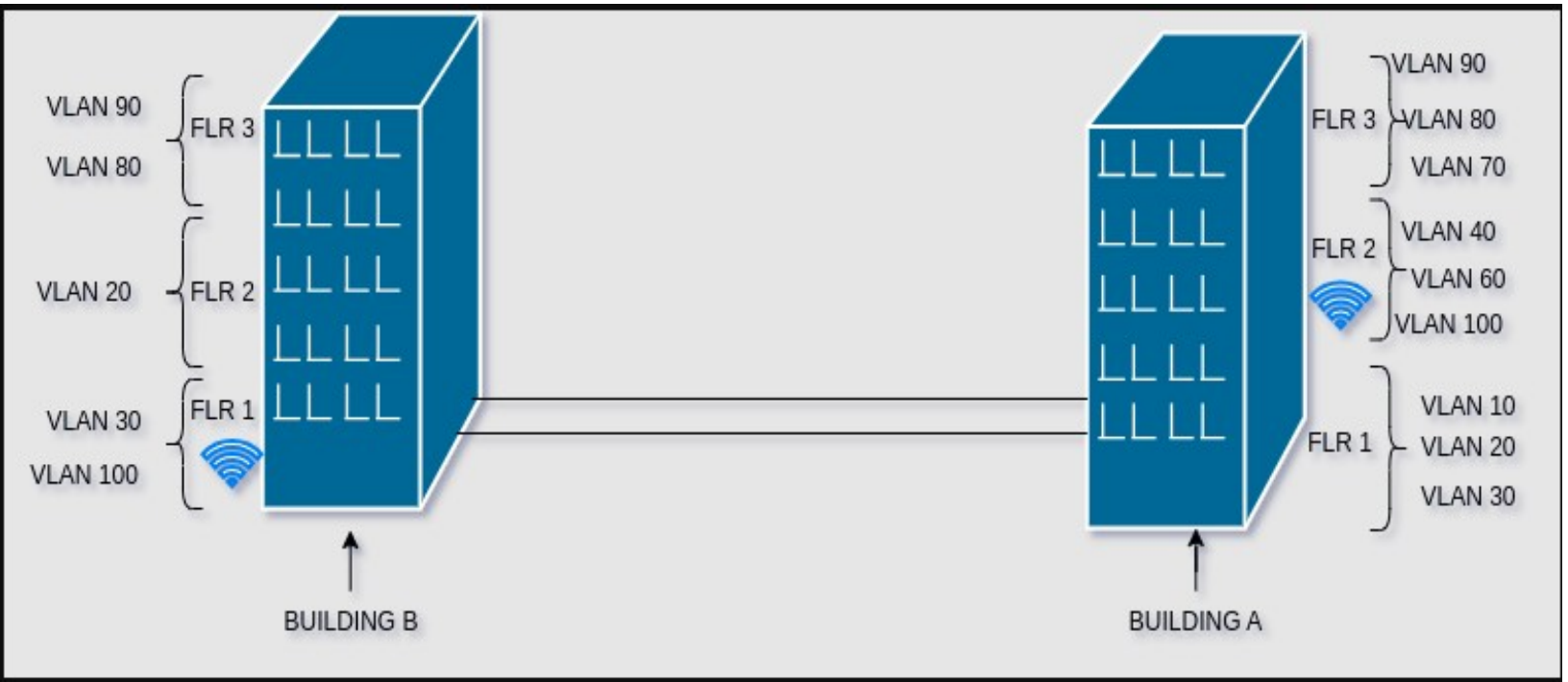
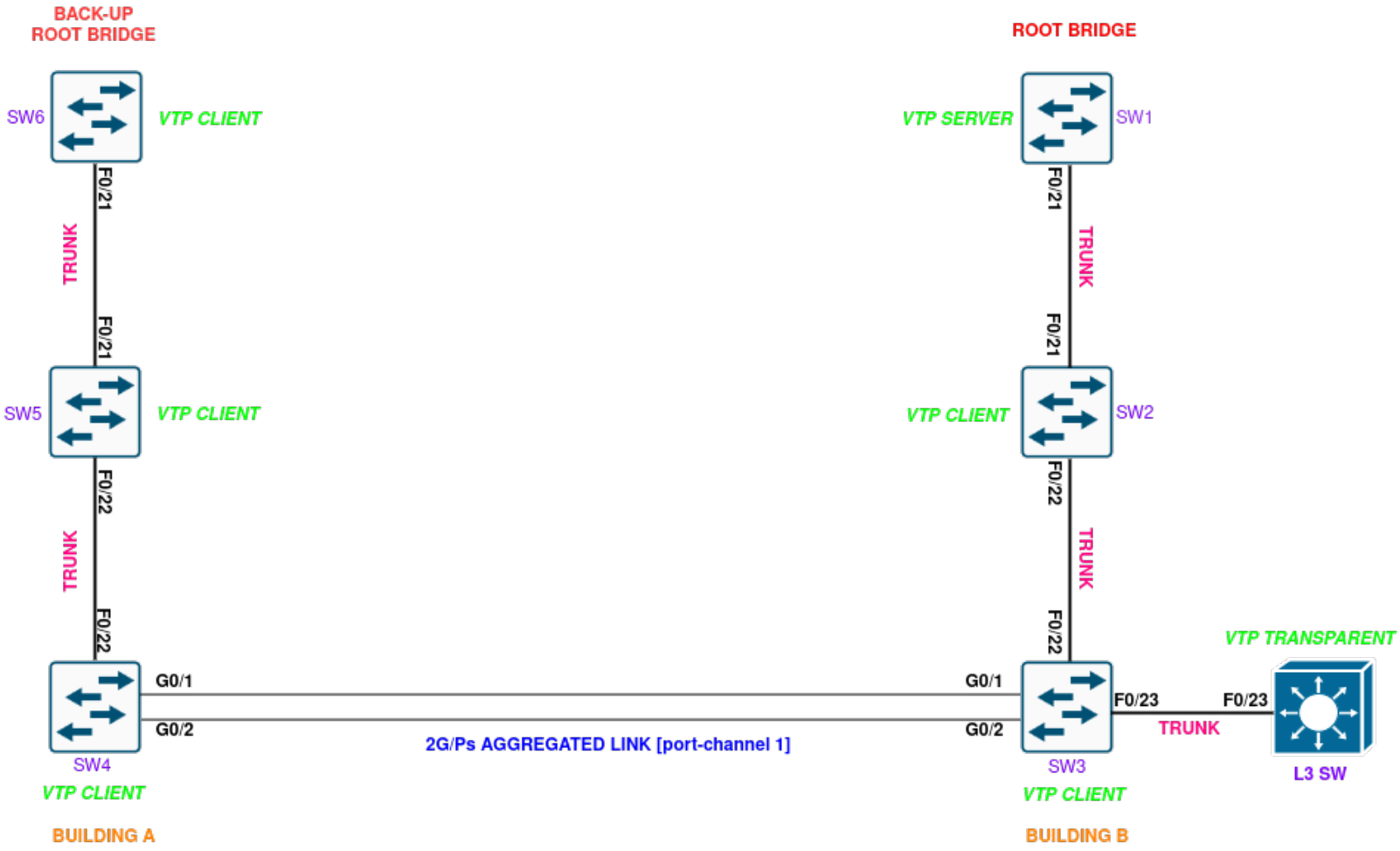
# IP SCHEME

| VLAN ID | VLAN NAME | SUBNET | DEFAULTGATEWAY |
|---|---|---|---|
| 10 | HR | 192.168.51.0/24 | 192.168.51.1 |
| 20 | ACCOUNTING | 192.168.22.0/24 | 192.168.22.1 |
| 30 | RECEPTION | 192.168.13.0/24 | 192.168.13.1 |
| 40 | EGINEERING | 192.168.14.0/24 | 192.168.14.1 |
| 60 | QUALITY-ASSURANCE | 192.168.38.0/24 | 192.168.38.1 |
| 70 | MARKETING | 192.168.70.0/24 | 192.168.70.1 |
| 80 | SALES | 192.168.18.0/24 | 192.168.18.1 |
| 90 | IT-SUPPORT | 192.168.15.0/24 | 192.168.15.1 |
| 100 | GUEST-WIRELESS | 192.168.10.0/24 | 192.168.10.1 |

# VLANS LAYOUT

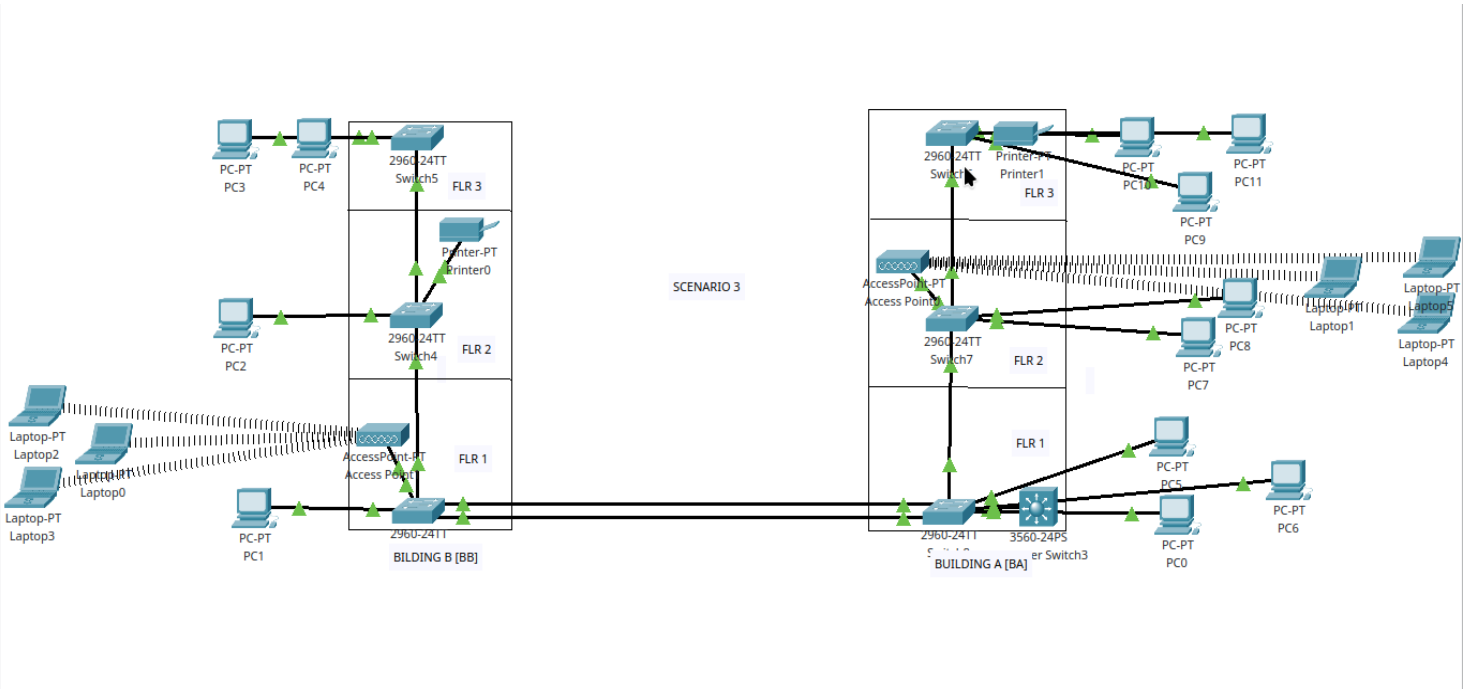


# LOGICAL CHAOS




https://draw.io/

# TOPOLOGY




*Built to scale, tested to servive.*

# VLAN TRUNKING PROTOCOL IMPLEMENTATION
# [VTP]

With this protocol, their three modes that Switchs can be configured through for automatic vlans propergation namely; **Server**, **Client** & lastly the **transparent** mode.

For this Topology, the **Multilayer** switch was configured as the a **VTP transparent** devices, to avoid over whelminng it with Vlans advertisements. The vlans were created localy so that **SVIs** could be configured with any failures.

Among the **Layer 2** switchs, one of them was configured as a **VTP server** and the Vlans **(Ids & Names)**, where localy created on this switch so that it could advertise correct vlan information to the other **Layer 2** switchs **(VTP clients).** As shown bellow**.**

```
MULTILAYER-SW#show vtp status
VTP Version capable             : 1 to 2
VTP version running             : 2
VTP Domain Name                 : cisco
VTP Pruning Mode                : Disabled
VTP Traps Generation            : Disabled
Device ID                       : 0060.2F36.0C00
Configuration last modified by 0.0.0.0 at 3-1-93 00:06:02

Feature VLAN :
--------------
VTP Operating Mode                : Transparent
Maximum VLANs supported locally   : 1005
Number of existing VLANs          : 14
Configuration Revision            : 0
MD5 digest                        : 0xF1 0x1E 0x5D 0x25 0x91 0x3F 0x93 0xB4
                                    0x8C 0x3E 0x1E 0x30 0x8C 0xB7 0x0C 0xB0

MULTILAYER-SW#
```

```
: 1 to 2
: 2
: cisco
: Disabled
: Disabled
: 0006.2A7A.4E00
).0.0.0 at 3-1-93 00:40:04

: Client
: 255
: 14
: 19
: 0x12 0x48 0x0F 0xF3 0xC8 0xFA 0x46 0xA9
  0x4A 0xCC 0x7E 0x51 0xBA 0x23 0x2D 0xDD
```

```
BA_FLR3-SW#show vtp status
VTP Version capable             : 1 to 2
VTP version running             : 2
VTP Domain Name                 : cisco
VTP Pruning Mode                : Disabled
VTP Traps Generation            : Disabled
Device ID                       : 0002.4A0E.C000
Configuration last modified by 0.0.0.0 at 3-1-93 00:40:04
Local updater ID is 192.168.15.2 on interface Vl90 (lowest numbered VLAN interface found)

Feature VLAN :
--------------
VTP Operating Mode                : Server
Maximum VLANs supported locally   : 255
Number of existing VLANs          : 14
Configuration Revision            : 19
MD5 digest                        : 0x12 0x48 0x0F 0xF3 0xC8 0xFA 0x46 0xA9
                                    0x4A 0xCC 0x7E 0x51 0xBA 0x23 0x2D 0xDD

BA_FLR3-SW#
```

**LAYER 3 SWITCH VIRTUAL INTERFECES**
**[SVIs]**

The **Layer 3** was switch configured for Inter-vlan Routing across the buildings, were by creating switch virtual interfaces **(SVI)** for each individual Vlan , assign it an IP address ending with a **.1 ,** and a subnet Mask in a **/24** scope. Then turned these interfaces up for the effective performance.

These **SVIs** were to be the **default gateways** for the vlans network addresses.
***Commands used.***
Enable
configure terminal
int vlan 90
description SVI for Vlan 90
ip address 192.168.15.1 255.255.255.0
no shut
exit

## SVIs

```
Vlan1                  unassigned      YES unset  administratively down down
Vlan10                 192.168.51.1    YES manual up                    up
Vlan20                 192.168.22.1    YES manual up                    up
Vlan30                 192.168.13.1    YES manual up                    up
Vlan40                 192.168.14.1    YES manual up                    up
Vlan60                 192.168.38.1    YES manual up                    up
Vlan70                 192.168.70.1    YES manual up                    up
Vlan80                 192.168.18.1    YES manual up                    up
Vlan90                 192.168.15.1    YES manual up                    up
Vlan100                192.168.10.1    YES manual up                    up
MULTILAYER-SW#
MULTILAYER-SW#
MULTILAYER-SW#
```

```
 no ip address
 description SVI for Vlan 10
 ip address 192.168.51.1 255.255.255.0
 description SVI for Vlan 20
 ip address 192.168.22.1 255.255.255.0
 description SVI for Vlan 30
 ip address 192.168.13.1 255.255.255.0
 description SVI for Vlan 40
 ip address 192.168.14.1 255.255.255.0
 description SVI for Vlan 60
 ip address 192.168.38.1 255.255.255.0
 description SVI for Vlan 70
 ip address 192.168.70.1 255.255.255.0
 description SVI for Vlan 80
 ip address 192.168.18.1 255.255.255.0
 description SVI for Vlan 90
 ip address 192.168.15.1 255.255.255.0
 description SVI for Vlan 100
 ip address 192.168.10.1 255.255.255.0
```

## DHCP CONFIGURATION

**Dynamic host configuration protocol** [DHCP] , is a network protocol where network hosts (user's devices), like laptops , smartphones and PCs are assigned Ips automatically once connected to that particular network. This is aimed to prevent IP conflicts as its error free, and also ease the work of a network administrator by automating his work of assigning Ips manually.

**DHCP**, was configured on the **Multilayer** switch of this topology, so that the hosts on each subnet can get assigned with an **IPV4** address, **Subnet mask**, the **default-gateway** address and the **DNS** address, all done automatically by the **Centralized** DHCP server which configured on the **Layer 3** switch.

To promote stability & availability of network devices such the Printers, **Layer 2 SVIs**, & servers etc. I had to ensure that they have **static** IPs though within the **DHCP** environment so that their IPs can't change even if power has gone off.
This was archieved after excluding a number of IP addresses of the subnets unless the Guest-Wireless were it wasn't necessary.

***Commands used.***

*enable*
*configure terminal*
*ip dhcp excluded-address 192.168.15.1 192.168.15.15*
*ip dhcp pool VLAN-90*
*network 192.168.15.0 255.255.255.0*
*default-router 192.168.15.1*
*domain-name vlan90.local*
*dns-server 192.168.15.8*
*exit*

**DHCP**

```
MULTILAYER-SW#show run | incl DHCP
MULTILAYER-SW#show run | incl dhcp
ip dhcp excluded-address 192.168.51.1 192.168.51.10
ip dhcp excluded-address 192.168.22.1 192.168.22.10
ip dhcp excluded-address 192.168.13.1 192.168.13.10
ip dhcp excluded-address 192.168.14.1 192.168.14.10
ip dhcp excluded-address 192.168.38.1 192.168.38.10
ip dhcp excluded-address 192.168.70.1 192.168.70.10
ip dhcp excluded-address 192.168.18.1 192.168.18.10
ip dhcp excluded-address 192.168.15.1 192.168.15.15
ip dhcp pool VLAN-10
ip dhcp pool VLAN-20
ip dhcp pool VLAN-30
ip dhcp pool VLAN-40
ip dhcp pool VLAN-60
ip dhcp pool VLAN-70
ip dhcp pool VLAN-90
ip dhcp pool VLAN-100
MULTILAYER-SW#
```

# TEST RESULTS

## SAME VLAN ON THE SAME BUILDING

### *VLAN 100 BB*

```
   Connection-specific DNS Suffix..: vlan100.local
   Link-local IPv6 Address.........: FE80::2D0:97FF:FEDE:86A8
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.10.2
   Subnet Mask.....................: 255.255.255.0
   Default Gateway.................: ::
                                     192.168.10.1

Bluetooth Connection:

   Connection-specific DNS Suffix..: vlan100.local
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0

C:\>ping 192.168.10.3

Pinging 192.168.10.3 with 32 bytes of data:

Reply from 192.168.10.3: bytes=32 time=72ms TTL=128
Reply from 192.168.10.3: bytes=32 time=32ms TTL=128
Reply from 192.168.10.3: bytes=32 time=42ms TTL=128
Reply from 192.168.10.3: bytes=32 time=40ms TTL=128

Ping statistics for 192.168.10.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 32ms, Maximum = 72ms, Average = 46ms

C:\>
```

### *VLAN 70 BA*

```
   Connection-specific DNS Suffix..: vlan70.local
   Link-local IPv6 Address.........: FE80::201:63FF:FEEC:1A54
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.70.12
   Subnet Mask.....................: 255.255.255.0
   Default Gateway.................: ::
                                     192.168.70.1

Bluetooth Connection:

   Connection-specific DNS Suffix..: vlan70.local
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0

C:\>ipconfig 192.168.70.11
Invalid Command.

C:\>ping 192.168.70.11

Pinging 192.168.70.11 with 32 bytes of data:

Reply from 192.168.70.11: bytes=32 time<1ms TTL=128
Reply from 192.168.70.11: bytes=32 time<1ms TTL=128
Reply from 192.168.70.11: bytes=32 time<1ms TTL=128
Reply from 192.168.70.11: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.70.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

# TEST RESULTS

## SAME VLAN ON  DIFFERENT BUILDING

### VLAN 100 BA – BB

```
   Connection-specific DNS Suffix..: vlan100.local
   Link-local IPv6 Address.........: FE80::2D0:97FF:FE2D:82B7
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.10.6
   Subnet Mask.....................: 255.255.255.0
   Default Gateway.................: ::
                                     192.168.10.1

Bluetooth Connection:

   Connection-specific DNS Suffix..: vlan100.local
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0

C:\>ping
C:\>ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Reply from 192.168.10.2: bytes=32 time=85ms TTL=128
Reply from 192.168.10.2: bytes=32 time=47ms TTL=128
Reply from 192.168.10.2: bytes=32 time=43ms TTL=128
Reply from 192.168.10.2: bytes=32 time=29ms TTL=128

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 29ms, Maximum = 85ms, Average = 51ms

C:\>
```

### VLAN 20 BB – BA

```
   Connection-specific DNS Suffix..: vlan20.local
   Link-local IPv6 Address.........: FE80::209:7CFF:FE1B:9793
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.22.12
   Subnet Mask.....................: 255.255.255.0
   Default Gateway.................: ::
                                     192.168.22.1

Bluetooth Connection:

   Connection-specific DNS Suffix..: vlan20.local
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0

C:\>ping 192.168.22.11

Pinging 192.168.22.11 with 32 bytes of data:

Reply from 192.168.22.11: bytes=32 time<1ms TTL=128
Reply from 192.168.22.11: bytes=32 time<1ms TTL=128
Reply from 192.168.22.11: bytes=32 time<1ms TTL=128
Reply from 192.168.22.11: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.22.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

# TEST RESULTS
## DIFFERENT VLANS ON THE SAME BUILDING

### VLAN 10 & 30 @ [BA]

```
FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..: vlan10.local
   Link-local IPv6 Address.........: FE80::290:21FF:FE92:5778
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.51.11
   Subnet Mask.....................: 255.255.255.0
   Default Gateway.................: ::
                                     192.168.51.1

Bluetooth Connection:

   Connection-specific DNS Suffix..: vlan10.local
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0

C:\>ping    192.168.13.11

Pinging 192.168.13.11 with 32 bytes of data:

Request timed out.
Reply from 192.168.13.11: bytes=32 time=1ms TTL=127
Reply from 192.168.13.11: bytes=32 time<1ms TTL=127
Reply from 192.168.13.11: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.13.11:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

### VLAN 100 & 90 @ [BB]

```
Wireless0 Connection:(default port)

   Connection-specific DNS Suffix..: vlan100.local
   Link-local IPv6 Address.........: FE80::2D0:97FF:FEDE:86A8
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.10.2
   Subnet Mask.....................: 255.255.255.0
   Default Gateway.................: ::
                                     192.168.10.1

Bluetooth Connection:

   Connection-specific DNS Suffix..: vlan100.local
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0

C:\>ping 192.168.15.17

Pinging 192.168.15.17 with 32 bytes of data:

Request timed out.
Reply from 192.168.15.17: bytes=32 time=38ms TTL=127
Reply from 192.168.15.17: bytes=32 time=28ms TTL=127
Reply from 192.168.15.17: bytes=32 time=24ms TTL=127

Ping statistics for 192.168.15.17:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 24ms, Maximum = 38ms, Average = 30ms

C:\>
```

# TEST RESULTS
## DIFFERENT VLANS ON DIFFERENT BUILDING

### VLAN 10 & 30 [BA - BB]

```
   Connection-specific DNS Suffix..: vlan10.local
   Link-local IPv6 Address.........: FE80::290:21FF:FE92:5778
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.51.11
   Subnet Mask.....................: 255.255.255.0
   Default Gateway.................: ::
                                     192.168.51.1

Bluetooth Connection:

   Connection-specific DNS Suffix..: vlan10.local
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0

C:\>
C:\>ping 192.168.13.12

Pinging 192.168.13.12 with 32 bytes of data:

Request timed out.
Reply from 192.168.13.12: bytes=32 time<1ms TTL=127
Reply from 192.168.13.12: bytes=32 time<1ms TTL=127
Reply from 192.168.13.12: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.13.12:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

### VLAN 80 & 90 [BB – BA]

```
FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..: vlan80.local
   Link-local IPv6 Address.........: FE80::202:4AFF:FE6C:4532
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.18.12
   Subnet Mask.....................: 255.255.255.0
   Default Gateway.................: ::
                                     192.168.18.1

Bluetooth Connection:

   Connection-specific DNS Suffix..: vlan80.local
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0

C:\>ping 192.168.15.16

Pinging 192.168.15.16 with 32 bytes of data:

Request timed out.
Reply from 192.168.15.16: bytes=32 time<1ms TTL=127
Reply from 192.168.15.16: bytes=32 time<1ms TTL=127
Reply from 192.168.15.16: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.15.16:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

# REDUNDANCY TESTING

A **redundant** link, ip an additional **route** or **path** between network devices like switches, that takes cover automatically in case the main link fials.

Two **gigabi**t links were installed these blocks, Runing from the switch at Floor 1 of building A to building B at ports gig 1 on gig 1 and gig2 conneted to gig2 of the other switch.

Top prevent **STP ,** from blocking one of these links, **Link aggregation** had to be configured. Where by after a 2GB/ps port-channel was formed in between these buildings for high speed transmission & efficiency of subnets traffic .

***Commands used on the two switches.***
*enable*
*configure terminal*
*int range G0/1-2*
*channel-group 1 mode active*
*exit*
*int port-channel 1*
*description 2Gb/ps Aggregated Link From BUILDING A - BUILDING B.*
*switchport mode trunk*
*switch trunk allowed vlan 10,20,30,40,60,70,80,90,10*
*exit*
*do show etherchannel summary*

*The Gig 1 to Gig 1* ***(primary path),*** *buildings connection was removed and ping tests were made on the same vlans and different vlans.*
*The* ***Redundant Link,*** *(Gig 2 on Gig 2) still worked correctly as the pings went on successfully.*

***Test Results***

```
C:\>ping 192.168.13.12

Pinging 192.168.13.12 with 32 bytes of data:

Reply from 192.168.13.12: bytes=32 time<1ms TTL=128
Reply from 192.168.13.12: bytes=32 time<1ms TTL=128
Reply from 192.168.13.12: bytes=32 time<1ms TTL=128
Reply from 192.168.13.12: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.13.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
C:\>ping 192.168.15.17

Pinging 192.168.15.17 with 32 bytes of data:

Request timed out.
Reply from 192.168.15.17: bytes=32 time<1ms TTL=127
Reply from 192.168.15.17: bytes=32 time<1ms TTL=127
Reply from 192.168.15.17: bytes=32 time=24ms TTL=127

Ping statistics for 192.168.15.17:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 24ms, Average = 8ms

C:\>
```

# SPANING TREE PROTOCOL
# [STP]

**STP,** is a protocol the automatically blocks **Redundant paths** once the **Primary path**(s) is still available until when it brakes and then the **redundant path** automatically acts as a **backup** path.

The **Layer 2** at the 3rd floor of Building **A** was selected manualy as the **Root Bridge** for the vlans, and this was because this floor is where the **Management** vlan hosts (Network Administrators) reside at this building.

***Commands used***
enable
conf t
spanning-tree vlan 1,10,20,30,40,60,70,80,90,100 root primary
exit

```
VLAN0010
  Spanning tree enabled protocol ieee
  Root ID    Priority    16394
             Address     00E0.B087.E090
             This bridge is the root
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    16394  (priority 16384 sys-id-ext 10)
             Address     00E0.B087.E090
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Fa0/21           Desg FWD 19        128.21   P2p
```

The **Layer 2** at the 3rd floor of Building **B** was selected manualy as the **Backup Root Bridge** for the vlans, and this was because this floor is where the **Management** vlan hosts (Network Administrators) reside at this building. This was archieved after using the following commands.

***Commands used***
enable
conf t
spanning-tree vlan 1,10,20,30,40,60,70,80,90,100 root secondary
exit

```
BB_FLR3-SW(config)#
BB_FLR3-SW(config)#
BB_FLR3-SW(config)#
BB_FLR3-SW(config)#spanning-tree vlan 1,10,20,30,40,60,70,80,90,100 root  secondary
BB_FLR3-SW(config)#
BB_FLR3-SW(config)#exit
```

This was aimed at ensuring that incase the primary **Root Bridge** at Building **A** fails to be reached by the other switches, then the **Backup Root Bridge** at Building **B** is selected automatically as the **Roo**t.

# VLANS IMPLEMANTATION
# scenario-[3]

**Designed, implemented,tested & documented**
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**END**

***powered by ;***